Domain: Business, Retail and E-commerce

Employee Attrition Intelligence: Predicting and Analyzing Employee Turnover

Abstract

In the business, retail, and e-commerce sectors, employee attrition has become one of the main problems that organizations have to deal with. Since high rates of employee turnover lead to higher recruitment and training expenses, they also have an impact on productivity, customer satisfaction, and the organization's overall performance. If companies understand the factors which lead to employee attrition and are able to predict which employees are likely to leave, they can then take proactive steps to improve employee retention.

The project known as "Employee Attrition Intelligence: Predicting and Analyzing Employee Turnover" is intended to create an intelligent predictive system based on machine learning methods in order to spot the employees who are in danger of leaving the organization. The system looks at a number of employee characteristics including age, job role, department, salary, years of experience, work-life balance, job satisfaction, overtime, promotion history, and performance ratings. The model identifies patterns and relationships which affect attrition by processing historical employee data.

The project includes performing data preprocessing, conducting exploratory data analysis (EDA), performing feature engineering, and implementing classification algorithms such as Logistic Regression, Decision Tree, Random Forest, and XGBoost to predict employee attrition. The models' performance is assessed using a number of metrics, including accuracy, precision, recall, F1-score, and ROC-AUC score, in order to determine the most reliable predictive model.

Together with its predictive capabilities, the system also gives analytical insights via interactive dashboards and visualizations, so that HR managers and business leaders can understand the main reasons for employee turnover. These insights aid decision-making based on data, which enables organisations to put into place specific retention strategies, boost employee satisfaction, optimise workforce planning and lower operational costs.

The project shows how data analytics and machine learning can be put into practice in the field of human resource management in business, in the retail sector, and in e-commerce. It helps to improve a stable, productive and engaged workforce together with greater organizational efficiency and long-term business growth by accurately forecasting employee attrition and identifying the main factors that influence it.

Keywords: Employee Attrition, Machine Learning, Predictive Analytics, Human Resource Analytics, Employee Retention, Business Intelligence, Retail, E-commerce, Classification, Data Visualization.